QuantumScope User Guide

Written for **QuantumScope 0.7.8** · Quantum Visualization Studio · MIT licensed

QuantumScope is an interactive studio for learning quantum physics: hydrogen orbitals, the experiments behind quantum mechanics, and quantum circuits, with built-in lessons that connect them. This guide shows how to use every part of it. To learn the physics itself, open 📖 **Learn** inside the app.

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1. Installing and updating

QuantumScope runs on **Windows** (64-bit), **macOS** (Apple Silicon) and **Ubuntu or Debian Linux** (64-bit). Download the installer for your system from the project's **Releases** page on GitHub (github.com/Onimee58/QuantumScope):

System	File	How to install
Windows	QuantumScope-...-Setup.exe	Run it and follow the steps.
macOS	QuantumScope-...-macOS.dmg	Open it and drag QuantumScope into Applications.
Ubuntu / Debian	quantumscope_..._amd64.deb	sudo apt install ./quantumscope_..._amd64.deb

First launch. The installers are not code-signed yet, so your system may warn you once:

- **Windows:** if SmartScreen appears, choose **More info → Run anyway**.
- **macOS:** if the app is blocked, open **System Settings → Privacy & Security** and choose **Open Anyway**.

Updating. Open ⓘ (top right) and choose **Check for updates...**, or use ⚙ → **Updates**. QuantumScope asks before it connects to GitHub. If a newer version exists, it shows the release notes; **Update now** downloads the installer, checks it against its published SHA-256 checksum, and installs it:

- **Windows:** the installer takes over, and QuantumScope closes.
- **Ubuntu:** enter your password when asked; QuantumScope waits for the installation and offers **Restart now**.
- **macOS:** the disk image opens; drag QuantumScope into Applications again.

Nothing is downloaded or installed without your confirmation, and a download that does not match its checksum is never installed.

2. A tour of the window

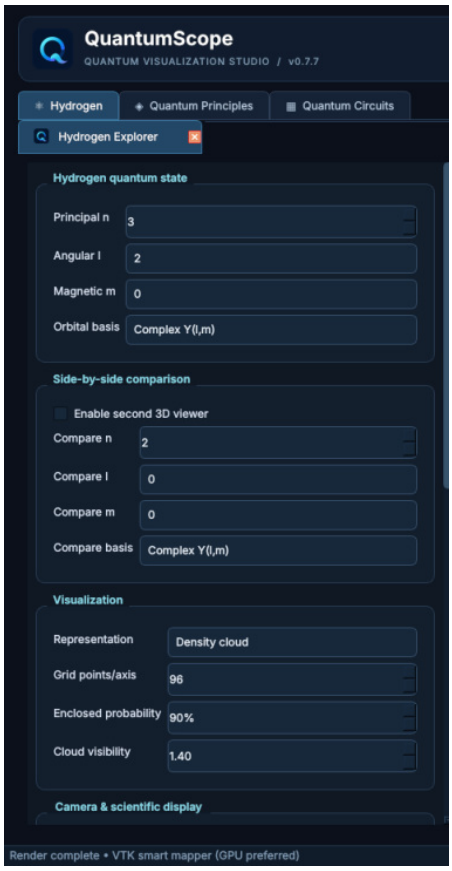


- **Areas:** the three tabs under the title, **Hydrogen**, **Quantum Principles** and **Quantum Circuits**, are the three areas of the app.
- **Experiment tabs:** each area holds experiment tabs. Open more with **+** on the right of the tab bar, or with **+ New Experiment** at the top. Every tab can be closed with its **×**. A closed tab is only hidden, so opening it again brings it back exactly as you left it. An area with no open tabs shows a **Start** page with its experiments as buttons.
- 📖 **Learn** opens the lessons for the current area (chapter 6).
- ⚙ opens **Settings** (chapter 7). ⓘ shows **About QuantumScope**, including **Check for updates…** and **User guide**, which opens this document.

When QuantumScope starts, its logo appears for a moment while the 3-D libraries load.

3. Hydrogen

The **Hydrogen Explorer** draws the hydrogen atom's orbitals in 3-D, exactly solved for the Schrödinger equation (with the proton's finite mass).



Choosing a state. Under **Hydrogen quantum state**, set **Principal n**, **Angular l** and **Magnetic m**. **Orbital basis** chooses between the complex orbitals **Y(l,m)** and the **Real cosine** or **Real sine** combinations (the familiar p_x, d_{xy} shapes). Press **Render state** to draw it; with **Update when changing state/mode** ticked, it redraws by itself.

How it is drawn. Under **Visualization**:

- **Representation: Density cloud** (a glowing cloud), **Probability enclosure** (a surface containing a chosen share of the probability), **Real(psi)** or **Imag(psi)**.
- **Enclosed probability** sets that share for the enclosure, for example 90%.
- **Grid points/axis** trades detail for speed, and **Cloud visibility** adjusts the cloud's brightness.
- **Color enclosure by wavefunction phase** colours the surface by the phase of ψ.

The view. Drag in the 3-D view to turn it, and scroll to zoom. Under **Camera & scientific display**, **View** jumps to **Perspective**, **+X**, **+Y** or **+Z**, and **Magnification** zooms. **Show axes and coordinates (a_mu)** adds axes in units of the (reduced-mass) Bohr radius. **Reset camera** returns to the start.

Comparing two orbitals. Tick **Enable second 3D viewer** under **Side-by-side comparison** and choose **Compare n**, **l**, **m** and **basis**. **Synchronize comparison cameras** keeps both views turned together.

The radial chart below the view shows $P(r) = r^2 \left|R_{nl}(r)\right|^2$, the probability of finding the electron at distance r.

Exporting. Export PNG saves the view (at the **PNG export scale**, 1–4×), and **Export grid (.npz)** saves the computed values for your own analysis.

Hydrogen Dynamics 2D and 3D

Open these with **+** in the Hydrogen area. They show a superposition of two states, **State A** and **State B**, evolving in time: the cloud moves because the two energies differ (a **quantum beat**).

- **State A probability** and **Initial phase** set the mixture.
- **Time / beat period** moves through one beat; **Play** animates it.
- The 2-D version shows a **Cross-section plane**; the 3-D version shows the full cloud.
- **Exports:** a frame as PNG, a full period as an animated GIF (with **GIF frames / period** and **GIF frames per second**), and, in 2-D, the slice as NPZ data.

4. Quantum Principles

Four hands-on experiments on the ideas behind quantum mechanics. They share some controls:

- **Bloch spheres: left-drag** sets a state (or, in Entanglement, aims a detector). **Right-drag**, or **Shift + left-drag** on a trackpad, turns the sphere to look from any side, and **double-click** resets the view.
- **Speed:** experiments that repeat measurements offer **Slow**, **Fast** and **Instant**, with **Stop**.

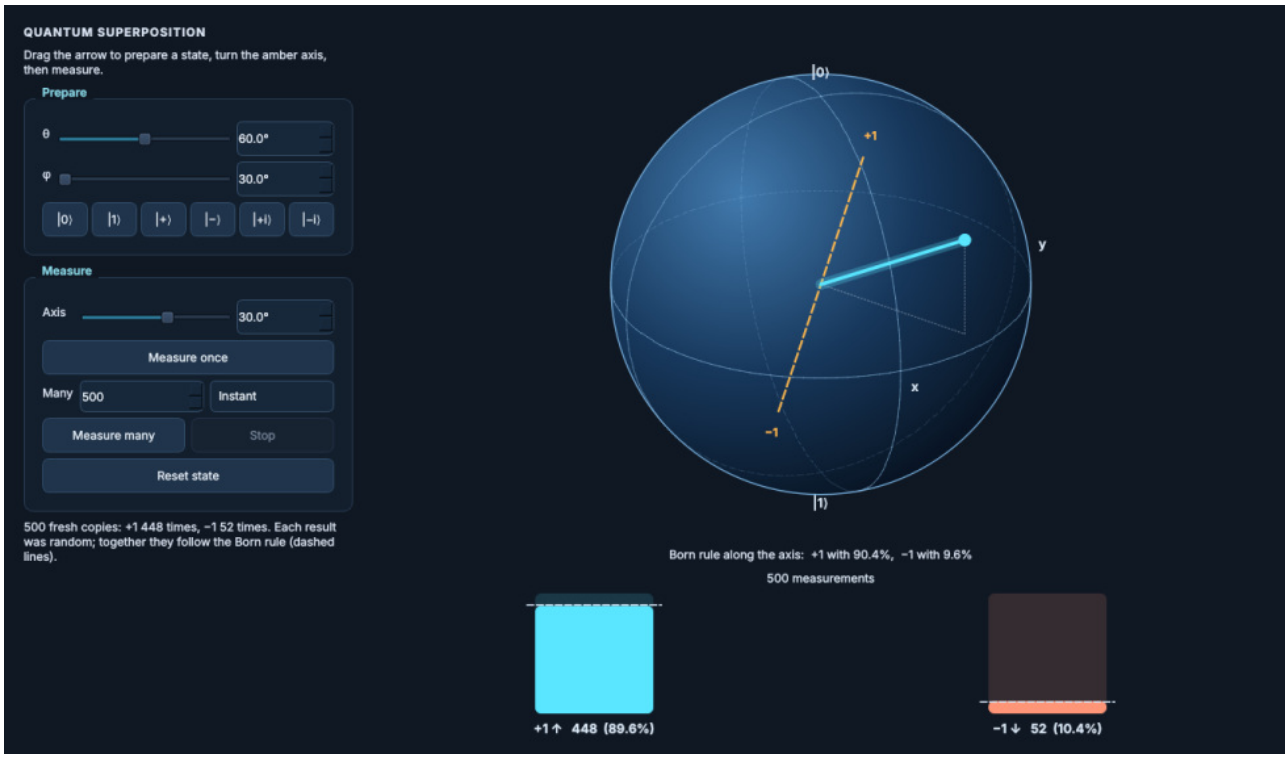
Double-Slit



Particles pass through two slits and land on a screen. **Fire particles** sends them one after another: each lands as a single dot, and the interference fringes appear only as the dots pile up. The chart compares the detections with the exact pattern.

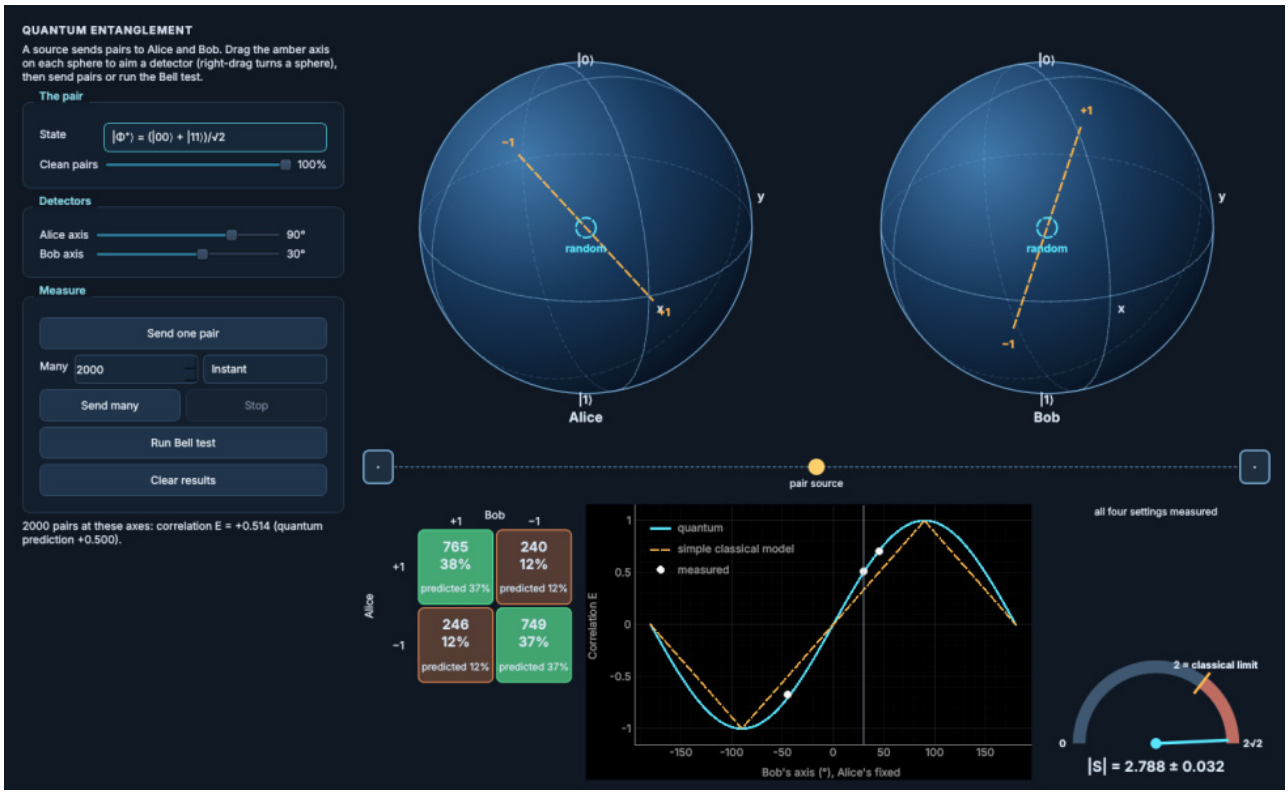
- **Wavelength** sets the light's colour; ultraviolet and infrared are shown in a labelled false colour.
- **Slit width**, **Slit separation**, **Screen distance** and **Screen half-width** set the geometry. You can also **drag the slits** in the drawing up or down to change their separation.
- **Which-path detectors at the slits** destroy the interference: the fringes give way to a single broad band.
- **Clear screen** starts again; **Export probability and detections (CSV)** saves the data.

Superposition



- **Prepare** a spin-½ state by **dragging the arrow**, with the **θ** and **φ** sliders, or with a preset: |0⟩, |1⟩, |+⟩, |−⟩, |+i⟩ or |−i⟩.
- **Measure: Axis** turns the amber measurement axis, whose ends are the results +1 and −1.
    - **Measure once:** the arrow jumps to +1 or −1 with a flash (the collapse). **Reset state** returns to the prepared state.
    - **Measure many:** fresh copies are measured one after another, and the bars grow toward the dashed **Born-rule** predictions.

Entanglement



A source sends pairs of qubits to Alice and Bob.

- **The pair:** choose a Bell state, or a **classically correlated** pair for comparison. **Clean pairs** adds noise.
- **Each sphere shows that qubit's own state:** for an entangled pair, **random**, since neither qubit has a state of its own.
- **Aim the detectors** by dragging the amber axis on each sphere, or with the **Alice axis** and **Bob axis** sliders.
- **Send one pair** follows a single pair: Alice measures, both qubits become definite, then Bob measures. No signal travels: on his own, Bob's results stay random.
- **Send many** fills the **grid** of joint results and the **correlation curve**, which compares the measurements with the quantum prediction and a simple classical model.
- **Run Bell test** measures the four standard detector settings and shows **|S|** with its uncertainty on the meter. Entangled pairs beat the classical limit of 2, up to 2√2 ≈ 2.83; noisy or classical pairs do not.

Teleportation



Alice sends Bob a qubit's state using a shared Bell pair and two classical bits.

- **The state to send:** a preset, or **Custom** with θ and φ; at step 1 you can also drag the arrow on q0.
- **Step by step: ◀ Back**, **Next ▶**, or **▶ Play**. The circuit at the top highlights each step. When Alice measures, her two bits travel to Bob along the classical channel before he applies his correction.
- **The measurement result:** choose one of the four, keep "unknown to Bob" for the average, or **Run once (random result)**.
- **Experiments:**
  - **Clean pairs** adds noise to the shared pair, lowering the fidelity.
  - **Bob applies the correction** can be unticked, to see why the classical bits are needed.
- **Teleport many** measures the average fidelity over many runs, against ½ (random), ⅔ (the best possible without entanglement) and 1.
- **Open in the circuit simulator** loads the same protocol as a circuit.

5. Quantum Circuits

The **Gate Circuit Simulator** builds and simulates circuits of up to 10 qubits, exactly.



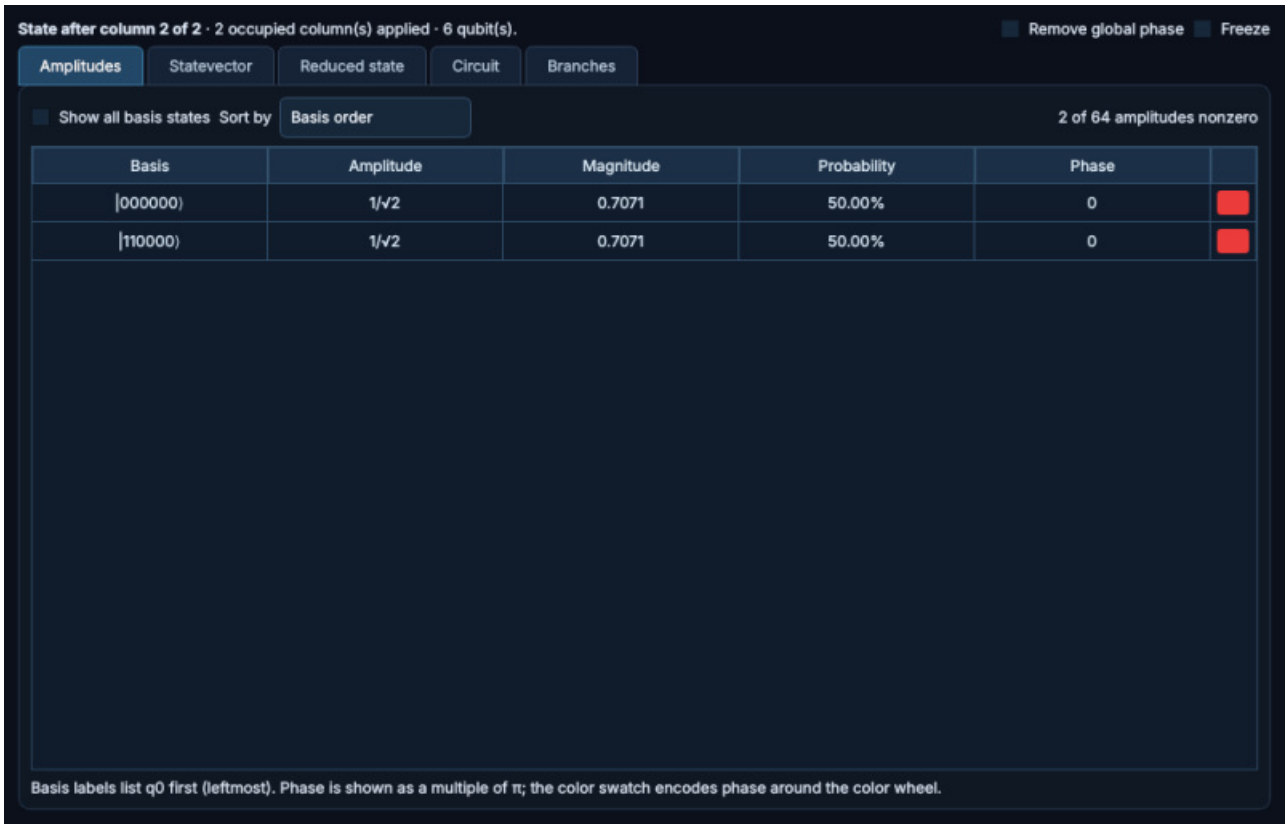
Building a circuit

- **Qubits** sets the number of qubits. **New gate angle** sets the angle given to new rotation gates (with **φ** and **λ** for the U gate).
- **Placing gates:** drag a gate from the palette on the left onto an empty cell. For a gate on several qubits (**CX**, **CZ**, **SWAP**, **CP**, **CCX**, **CSWAP**), drop it on its first qubit, then click the other qubits in the same column.
- **Editing:** click a placed gate to select it. **Double-click** it to edit its angles (or a noise box's strength); double-clicking a grouped **block** expands it. **Right-click** removes a gate.
- **Qubit labels:** right-click a qubit's name on the left to give it a label, such as *data* or *Alice*.
- **Undo** and **Redo** step through changes. **Clear** empties the circuit.
- **Zooming: − Zoom** and **+ Zoom**, **Fit Circuit** and **Reset Zoom**.
- **The palette:**
  - **Gates:** H, X, Y, Z, S, T, SDG, TDG, SX, P, RX, RY, RZ, U, CX, CZ, SWAP, CP, CCX, CSWAP and M (a final measurement).
  - **Classical: M→c**, a measurement in the middle of the circuit into a classical bit, and **Reset**.
  - **Noise: Bit flip**, **Phase flip**, **Depolarize**, **Amp. damp** and **Dephase**, which can be placed like gates.
  - **Double-click** any gate or noise item in the palette for an explanation of what it does.

Running and reading results

- **Playback: ◀, ▶ Play** and ▶ step through the circuit column by column; the probabilities and Bloch spheres show the state at that point.
- **Probabilities:** the chart shows each basis state's probability, with the theory in cyan. The line underneath lists them.
- **Measurement shots:** set the number of shots and press **Run shots** to sample results as an experiment would.
- **State Inspector...** opens a window with the **Amplitudes**, **Statevector**, **Reduced state** (each

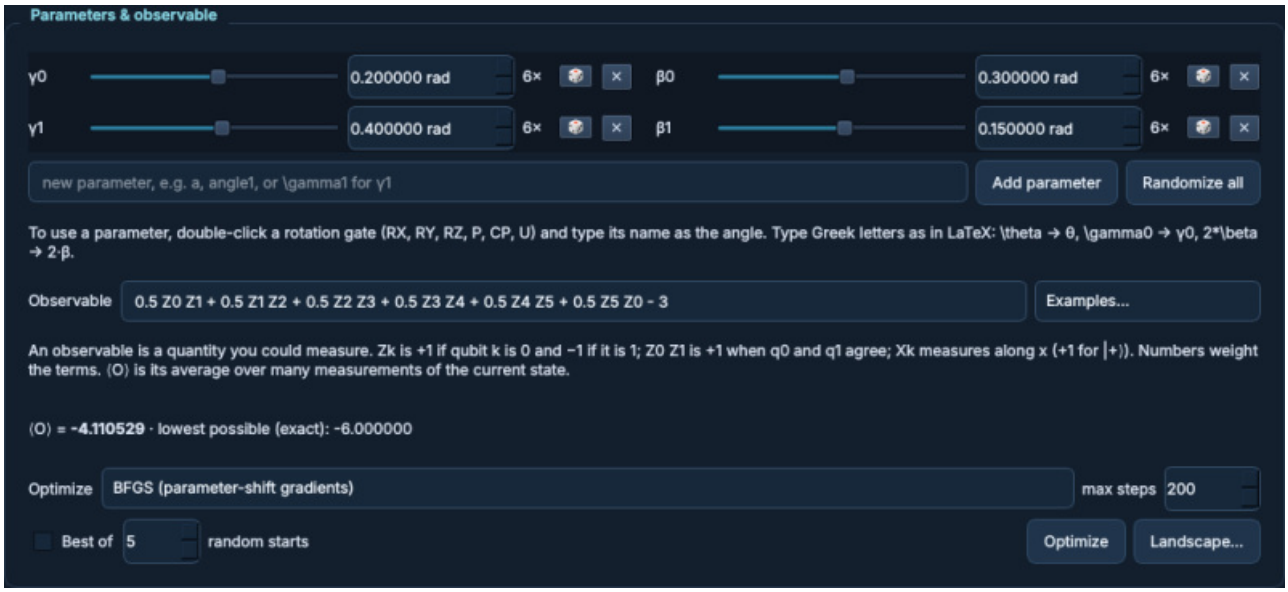
qubit's own state), **Circuit** statistics and, for circuits with mid-circuit measurements, **Branches**.



Examples, algorithms and checking

- **Examples…** loads ready-made circuits: the Bell pair, GHZ, adders and comparators, teleportation, superdense coding, and the bit-flip, phase-flip and Shor codes.
- **Algorithms…** loads Deutsch, Deutsch–Jozsa, Bernstein–Vazirani, Grover search, the quantum Fourier transform, phase estimation, VQE and QAOA. A bar above the circuit names the example; **Change parameters…** adjusts it, and **Learn more** opens its lesson.
- **Verify / truth table** checks the circuit against its specification: for an example, every required input; for your own circuit, a truth table of basis-state inputs.
- **Plot logical error rate** appears for error-correction examples, and **Show the graph** for QAOA.
- **Noise: off** opens the noise settings: presets, error rates, idle noise and readout errors. With noise on, the simulation uses density matrices.

Parameters and optimization



Press **Parameters** to show the panel.

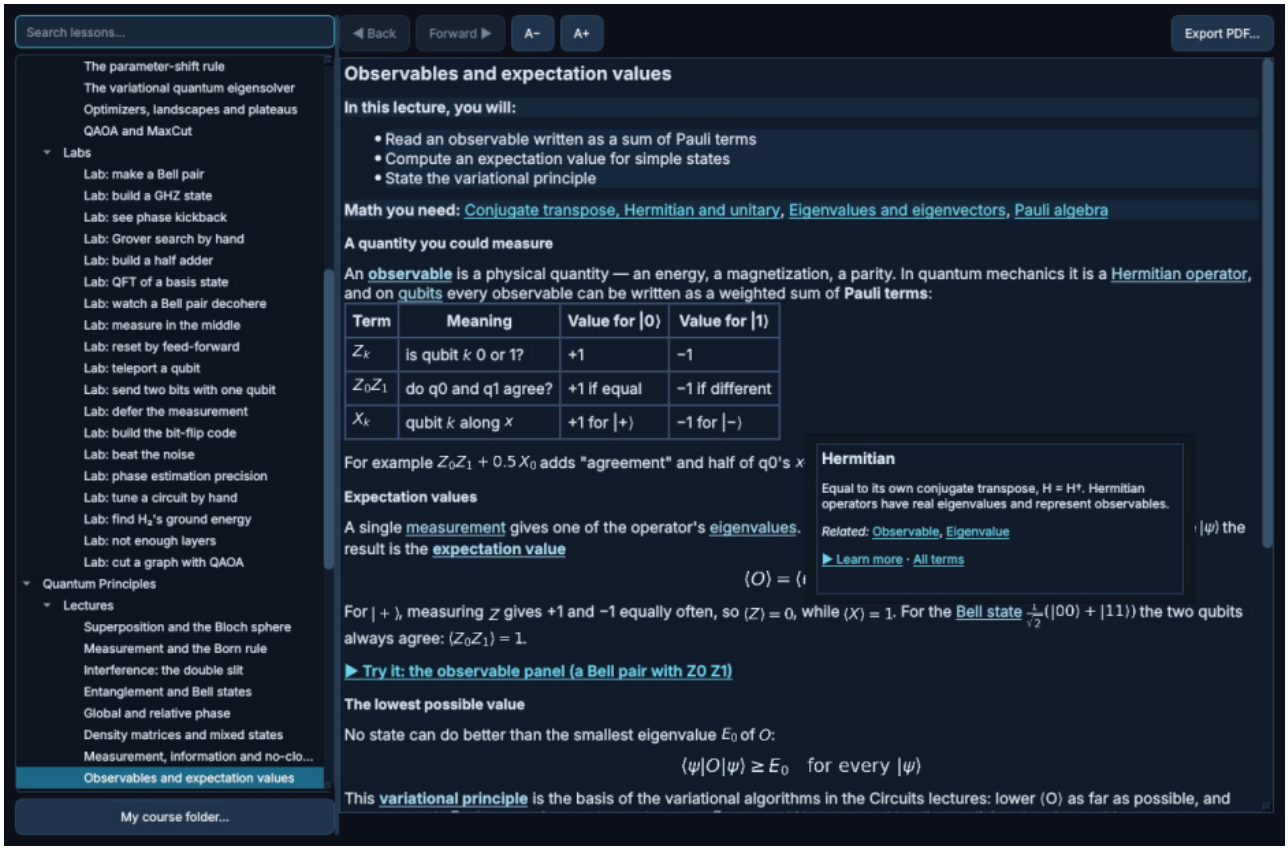
- **Parameters:** add a name with **Add parameter** (type \theta for θ, \gamma0 for γ0). Then double-click a rotation gate and type the name, or a multiple like 2*\beta, as its angle.
    - **Sliders** change parameters live. On wide windows they show in two columns.
    - 🎲 randomizes one parameter, and **Randomize all** randomizes every one.
- **Observable:** type a weighted sum of Pauli terms, such as Z0 Z1 + 0.5 X0, or pick one from **Examples...**. The panel shows its expectation value ⟨O⟩ and the lowest possible value.
- **Optimize** lowers ⟨O⟩ with the chosen method (**BFGS**, **Gradient descent**, **COBYLA** or **Nelder–Mead**). **Best of N random starts** also tries random starting points and keeps the best.
- **Landscape...** draws ⟨O⟩ over two parameters, with the optimizer's path.

Files and tools

- **File: Open .qscope...**, **Open recent**, **Save .qscope...**, **Import OpenQASM 2...**, **Export OpenQASM 2...** and **Export PDF...** (the circuit diagram, optionally with a probability report).
- **Tools: Simplify circuit...** and **Check equivalence with a saved circuit...**.
- **The QASM Converter** (open it with **+**) converts between OpenQASM 2 and QuantumScope's .qscope format.
- **The Error-correction explorer** (open it with **+**) compares how the error-correcting codes perform as noise grows.
- **Autosave:** QuantumScope saves your circuit as you work and offers to recover it after a crash (see ⚙ → **Files**).

6. Learn

📖 **Learn** opens QuantumScope's course: lectures and labs for every area, plus **Mathematics**.

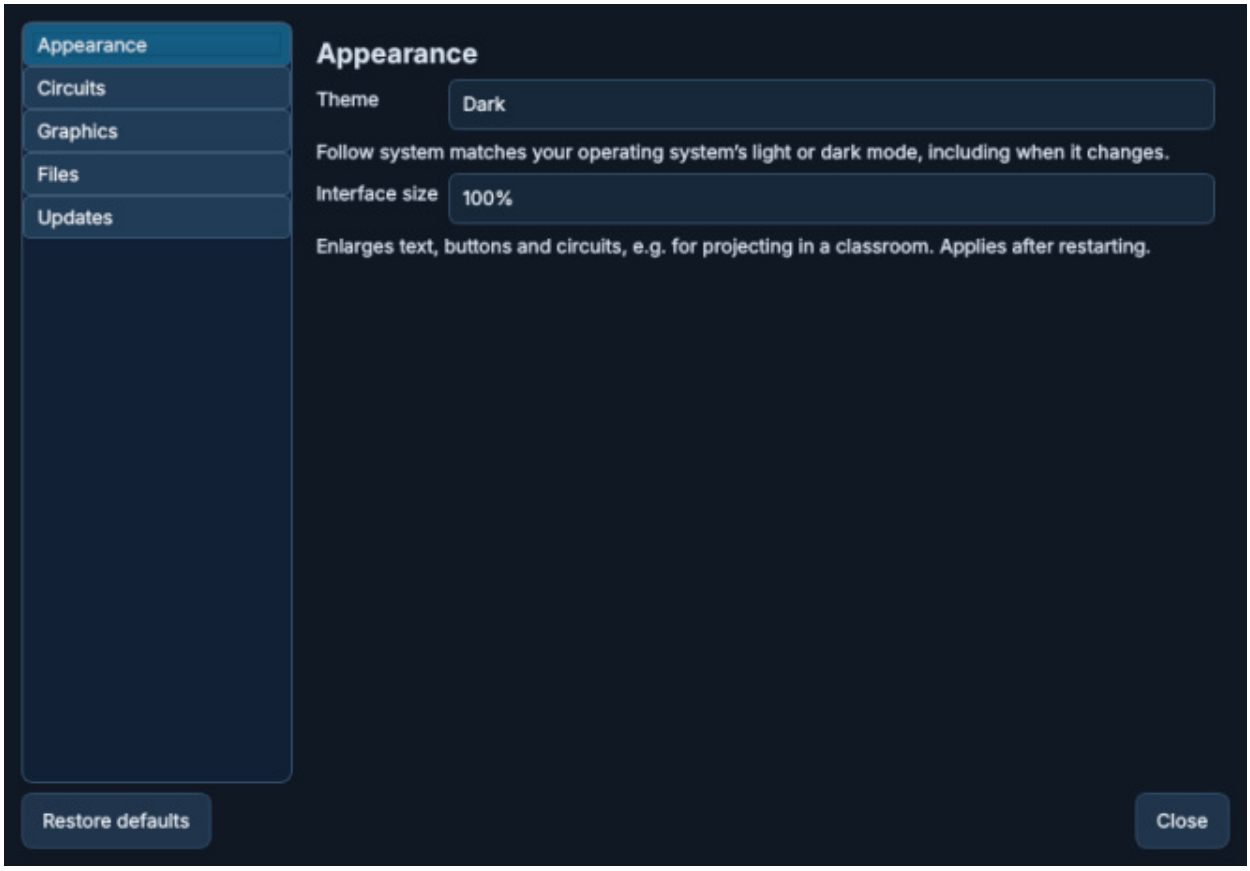


- **Contents:** lessons are grouped by area: **Quantum Circuits**, **Quantum Principles**, **Hydrogen** and **Mathematics**. **Search lessons...** finds any lesson by its words.
- **Lectures** explain a topic. **Math you need** at the top links to the Mathematics lessons it builds on.
- **Try it** links open the matching experiment or circuit in the main window.
- **Quizzes** check your understanding: click an answer to see whether it is right, and why.
- **Labs** give step-by-step tasks in the circuit simulator and check each step as you complete it.
- **The glossary:** underlined terms, such as *Hermitian*, show a short definition when clicked, with related terms and **Learn more**. **Glossary** in the contents lists all of them.
- **Reading:** **◀ Back** and **Forward ▶** move through the lessons you have visited, and **A−** and **A+** change the text size.
- **Export PDF...** saves the lesson as a handout, ending with the terms it uses.

For teachers: My course. My course folder... adds your own lessons and labs, written in the same simple format. They appear under **My course**, with glossary links and **Math you need** like the built-in ones. The file LEARN.md in the project describes the format.

7. Settings

⚙ opens **Settings**. Changes apply at once, except where noted; **Restart now** restarts when needed, and **Restore defaults** resets everything (your recent files are kept).



- **Appearance: Theme** (**Dark**, **Light** or **Follow system**) and **Interface size** (100–175%, useful for projecting in a classroom; applies after a restart).
- **Circuits:** defaults for new circuits:
    - **Qubits in a new circuit** and **Measurement shots**;
    - **Gate angles** in degrees or radians, and **Bit order in labels**, q0 leftmost (textbook) or rightmost (Qiskit);
    - **Amplitudes** shown exactly (1/√2) or as decimals, with **Decimal places**;
    - **Ask before clearing a circuit**.
- **Graphics: Graphics device** (**Automatic**, **Integrated graphics**, or **NVIDIA (PRIME offloading)** on Linux laptops with two GPUs; applies after a restart) and **3D volume rendering** (**Auto**, **GPU** or **CPU**). **Refresh** shows which graphics renderer is in use.
- **Files:**
    - **Save and export to:** the default folder (**Browse…** or **Use Documents**);
    - **Autosave**;
    - **Probability report in PDF exports:** ask each time, always, or never;
    - **Recent circuits**, with **Clear recent list**.
- **Updates:** the current version and **Check for updates…**.

8. Troubleshooting

The 3-D view is blank, black or very slow. Open ⚙ → **Graphics**. Try **3D volume rendering: CPU**, which works on any computer. On a laptop with two graphics chips, try the other **Graphics device**, then restart. **Refresh** shows which renderer is actually in use; include it if you report a problem.

Everything is too small on a projector or a high-resolution screen. Set ⚙ → **Appearance** → **Interface size** to 125–175% and restart.

Basis states are written in the opposite order to my textbook. Change ⚙ → **Circuits** → **Bit order in labels**. Only the labels change; the physics is identical.

QuantumScope closed unexpectedly. When it starts again, it offers to recover your autosaved circuit.

Check for updates says it cannot connect. Check your internet connection and try again later. GitHub also limits how often it can be asked, so wait a while if you have checked many times. You can always download the latest release from the Releases page.

Reporting a problem. Open an issue on GitHub (github.com/Onimee58/QuantumScope). Describe what you did, what happened, your system, and the version shown in ⓘ.

9. Mouse and keyboard reference

Where	Action	Result
Circuit	drag from the palette	place a gate
Circuit	click a gate / double-click it	select it / edit its angles
Circuit	right-click a qubit's name	give the qubit a label
Circuit	right-click a gate	remove it
Circuit	click other qubits after dropping a multi-qubit gate	choose its other qubits
Palette	double-click a gate or noise item	explain it
Bloch sphere	left-drag	set the state (Entanglement: aim the detector)
Bloch sphere	right-drag, or Shift + left-drag	turn the sphere
Bloch sphere	double-click	reset the view
Double-Slit	drag the slits	change their separation
Hydrogen 3-D view	drag / scroll	turn / zoom
Learn	click an underlined term	show its definition
Any tab	× on the tab	close it (it reopens as you left it)

QuantumScope is open-source software (MIT licence) by Md. Saif Hassan Onim, Southern Methodist University. Its simulations are idealized for teaching.